

Seth Baird

Mobile : (469) 486-2236

LinkedIn : seth-baird-8b68b9243

Email : sethbaird01@gmail.com

Github : sethbaird01

WORK EXPERIENCE

- **SMS InfoComm (Wistron Corporation)**

Dallas, TX

NVIDIA Server Repair Technician

May 2024 - August 2024

- **System Repair and Troubleshooting** Diagnosed, refurbished, repaired, and built NVIDIA HGX 8-GPU units.
- **Hardware Validation and Analysis** Used proprietary diagnostic tools to gain insight into the operational state of motherboard subsystems such as power delivery, signal integrity, and FPGA.
- **Servers and Information Technology** Daily activities depended on the use of Linux, SSH, tmux, Microsoft Office, as well as strong communication and note-taking skills.

- **Jawa.gg**

Remote

Verified Seller

October 2021 - August 2023

- **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
- **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
- **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.

INVOLVEMENT

- **Purdue Electric Racing - STM32 Software Team**

- Wrote multiple HALs (Hardware abstraction layers) to simplify the codebase and make development easier
- Reworked the CAN communication backend to use two busses simultaneously for new high-throughput motors
- Utilized testing equipment such as CANable to verify interoperability of new and existing systems

SKILLS AND PROJECTS

- **Microcontroller, FPGA, and IoT**

- Implemented channel scan into a superheterodyne FM radio using regression and a local maximum search algorithm
- Leveraged Solidworks, OpenSCAD, and a Raspberry Pi Pico running Rust to create a replica arcade game controller
- Added smart capability to a ceiling fan using an ESP8266 Wi-Fi microcontroller (Arduino C) and the Blynk API
- Designed and simulated logic circuits and HDL code for Altera FPGAs in Quartus.

- **Software Development**

- Developed an android app that integrated with a car's engine computer to provide data and insights to the driver
- Completed a 2 semester long course on data structures at a vocational school before enrollment in university
- Studied OOP in Java at home and vocational school beginning in 2018, completing 3 courses in the subject
- Created over 20 proof-of-concept iOS and Android apps to gain familiarity with Swift and Kotlin
- Leveraged Node.js, Discord.js, and Dotenv to develop an audio capable bot written in Typescript

- **Linux Server Administration**

- Designed custom bash scripts utilizing cron to automate ZFS dataset replication and other maintenance tasks
- Administrated a network of KVM virtual machines and Linux containers (LXC) under the Proxmox hypervisor
- Configured a DNS server with domain filtering to block advertisements and malicious sites for all LAN clients
- Built a pair of multipurpose storage servers configured to make bidirectional encrypted backups over VPN

EDUCATION

- **Purdue University**

West Lafayette, Indiana

In progress - Anticipated Graduation May 2027

GPA 3.01

Major in Computer Engineering Technology Minor in Computer and Information Technology

- **Memorial High School**

Frisco, Texas

Graduated May 2023

Unweighted GPA 3.81

Vocational Education: CS3: Data Structures and Algorithms, Mobile Application Development